

Talk Like a Structured Graph? Structural Statistics Help Small LLMs Mainly by Stating the Answer

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Abstract

Language models are increasingly used to reason over structured data, yet a graph written out as text is still hard for them to read, and their accuracy depends on how it is written. Graph neural networks benefit from explicit structural features, but it is unclear whether writing the same statistics into a textual graph helps a frozen language model execute graph queries. We prepend node degree, local clustering coefficients, and random-walk return probabilities to GraphQA’s incident encoding and evaluate Qwen3-1.7B and Qwen3-4B, with and without thinking, on paired graphs. Stated degrees improve node-degree accuracy when unaided counting is unreliable, but reduce it for a model that already counts accurately. This benefit does not establish improved graph reading: degrees directly supply the queried value, and responses often use a lookup rather than the incident edges. Clustering and return probabilities, which do not directly state the answer, yield smaller effects that vary by model and task, with no significant gain in the joint accuracy of degree and neighbor-set queries. A component-count control also changes how a model reads the graph despite conveying little new information on these graphs. Explicit statistics can help by supplying answers or changing how a model uses the prompt, but do not consistently improve graph-query execution.¹

1 Introduction

Structural features and positional encodings improve graph neural networks and graph Transformers by supplying information that message passing may not readily recover (Dwivedi et al., 2022; Ram-pásek et al., 2022). A language model given a graph as text must instead extract structure from node and edge descriptions. GraphQA reports difficulty on basic graph tasks and shows that performance depends on the encoding, task, and graph structure;

NLGraph likewise finds that prompting becomes less effective on more complex problems (Fatemi et al., 2024; Wang et al., 2023). We ask: *when a black-box language model reads a textual graph, do explicit structural statistics improve execution of graph queries?*

GraphQA varies how graphs are serialized (Fatemi et al., 2024); other work supplies learned graph tokens (Perozzi et al., 2024) or examines structural prompts for node classification on text-attributed graphs (Huang et al., 2024). These studies leave a narrower question open: whether pre-computed statistics help a model answer graph queries when the textual edge list and question are fixed. A degree sentence can itself contain the answer to a degree query, so an accuracy gain need not indicate better use of the edges.

We test this question by placing a primer of degree, local clustering coefficients, or random-walk return probabilities before GraphQA’s incident encoding (Figure 1). We evaluate Qwen3-1.7B and Qwen3-4B, with and without thinking, on paired graph-question prompts across six GraphQA tasks. The graph, encoding, and question remain fixed across primer conditions. We also test the combined statistics, a component-count sentence, and a structure-free filler. A graph-blind solver measures what the primer text alone makes available; paired comparisons, joint degree and neighbor-set correctness, and checks of response claims against the graph provide further evidence about how answers are obtained.

Our contribution is a controlled evaluation of these statistics as textual primers for graph queries, with controls for answer supply and the effects of added text. This framework examines whether a change in accuracy reflects information given directly by a primer or improved use of the graph’s incident encoding.

¹<https://github.com/GalBrk/GraphTalk>

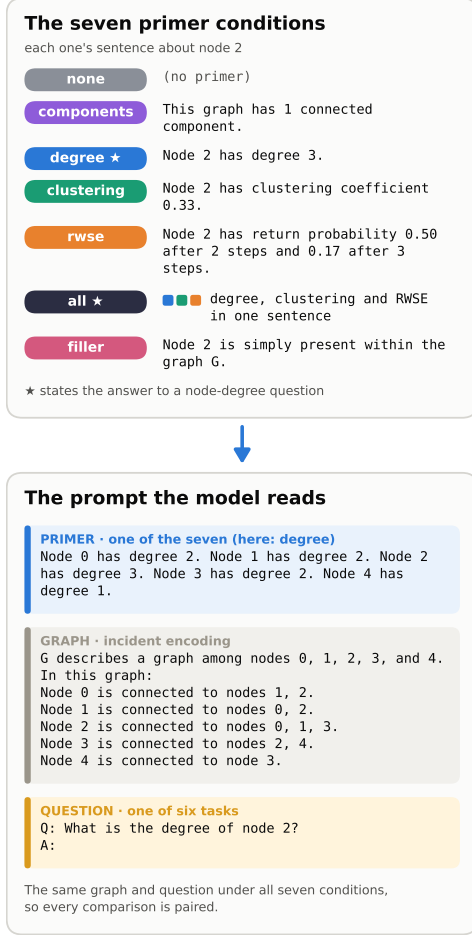


Figure 1: How a primer is used. Top: the seven conditions, each with its sentence about node 2 of a five-node example (the study’s graphs have 40 nodes); ★ marks the two that state a node-degree answer. Bottom: the prompt the model reads, one primer followed by GraphQA’s incident encoding and the question.

2 Related Work

Graph representations for LLMs. GraphQA studies how textual encodings, prompts, and graph structure affect graph-query accuracy (Fatemi et al., 2024); NLGraph evaluates natural-language graph problems and prompting methods (Wang et al., 2023). Other approaches supply learned graph tokens (Perozzi et al., 2024), motif descriptions (Das et al., 2024), or structural context for node classification on text-attributed graphs (Huang et al., 2024). Our experiment holds the incident encoding and question fixed while varying named statistics in plain text, including whether a statistic supplies the queried answer.

Controls and response evidence. Added context, input length, and formatting can change model answers (Shi et al., 2023; Levy et al., 2024; Sclar

et al., 2024). A successful partial-input baseline exposes a shortcut, but an unsuccessful one cannot exclude other shortcuts (Feng et al., 2019). Generated explanations may also misrepresent how an answer was reached (Turpin et al., 2023). We therefore compare primers with a structure-free filler, measure what a graph-blind solver can recover, and check stated graph claims against the actual edges. These diagnostics provide complementary evidence rather than direct access to the model’s internal computation.

3 Method

Models and graphs. We evaluate Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025), each with thinking off (P) and on (T), using zero-shot greedy decoding and one response per prompt (budgets in Appendix A). A pilot on published GraphQA graphs left larger models near ceiling (Appendix C). We therefore use 40-node independent-edge graphs $G(n, p)$ (Gilbert, 1959) at $p \in \{.10, .20, .35, .50\}$, extending node-degree and edge-existence tests to $p \in \{.65, .75, .85\}$. Every condition and arm receives the same 100 graphs per task and density, allowing paired comparisons.

Tasks and primers. We retain GraphQA’s incident encoding and six tasks (Figure 1). Node degree, connected nodes, edge count, and edge existence have varying answers. We score connected nodes primarily by exact match of the predicted and true neighbor sets; set-F1 is a secondary measure that gives partial credit. Node count is always 40, and every sampled graph contains a cycle. Cycle check therefore cannot measure whether a model distinguishes cyclic from acyclic graphs; we use its responses to assess the support given for a “yes” answer.

Before the encoding, we place either no primer; one sentence per node stating its degree, local clustering coefficient, or two- and three-step random-walk return probabilities (a truncated RWSE; Dwivedi et al., 2022; Rampásek et al., 2022); or a sentence combining all three statistics. Controls give the component count or name each node without stating a statistic (filler). The filler is similar in length to clustering but does not match every primer (Appendix Table 3). A rule-based graph-blind solver tests what can be answered from primer text alone.

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	60.50	+7.5 ^f	+3.0	+4.0	+10.2 ^f	+1.8	-0.5
	1.7T	76.25	+8.8 ^f	+1.5 ^f	+2.5 ^f	+12.2 ^f	-1.0	-3.0
	4P	99.25	-6.5 ^f	-0.5	-2.2	-8.0 ^f	+0.2	-0.5
	4T	97.00	+2.5	+1.8	-1.2 ^f	+0.5	+0.2	+2.0
Neighbors	1.7P	72.50	+3.5 ^f	+1.2 ^f	+1.0 ^f	+2.0 ^f	+0.8 ^f	-6.2
	1.7T	78.75	+6.2	+2.0	+0.2	+6.2	+2.5	+2.5
	4P	86.00	-3.0 ^f	+1.0 ^f	+0.0 ^f	-2.8 ^f	+9.2 ^f	-12.5
	4T	95.50	+0.5	-0.8	+0.5	-1.2	+0.5	+1.2
Edge count	1.7P	1.25	+3.0 [†]	-0.5 [†]	-0.8 [†]	+0.8 [†]	-1.0 [†]	-0.5 [†]
	1.7T	15.75	-3.2 [†]	-1.2 [†]	-3.2 [†]	+5.0 [†]	-0.8 [†]	-1.5 [†]
	4P	2.00	+27.2 ^f	-1.2 ^f	-1.5 ^f	+3.8	-1.2 ^f	+1.2
	4T	20.00	+8.5 [†]	+24.8 [†]	+10.2 [†]	+1.8 [†]	+24.8 [†]	+12.2 [†]
Edge exists	1.7P	70.50	+4.0 ^f	+3.8 ^f	+5.0 ^f	+11.5 ^f	-8.2 ^f	-15.8
	1.7T	99.00	-2.5	-1.0	-1.2	-1.8	-0.8	-0.5
	4P	90.50	-4.8	+4.0 ^f	-10.0	-10.0	-2.8 ^f	-7.0
	4T	99.75	+0.0	-0.8	-1.2	-1.0	-0.5	+0.0

Table 1: Primary result: correct share without a primer (% of all prompts) and paired change under each primer (points), on the same 400 graphs per row ($p \leq .50$). Shaded: the primer supplies the degree answer or every term needed to compute edge count. Bold: $q < .05$ against no primer (BH within arm, task and control); ^f: $q < .05$ against filler; [†]: at least 15% truncation on either side, and no significance marks. Intervals and constant-answer tasks: Table 4; truncation: Table 7; balanced accuracy: Table 5. P/T: plain/thinking.

Outcomes and inference. Each response is classified as correct, wrong, or truncated; accuracy uses all prompts as its denominator. The primary effect is the paired change in accuracy, in percentage points, against no primer and against filler on the 400 main-sweep graphs per task (Table 1). We use exact McNemar tests (McNemar, 1947), bootstrap intervals over graphs, and Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) within each arm–task–control family. Cells with at least 15% truncation in either condition are flagged and excluded from pooled summaries. We also report balanced accuracy for edge existence because its positive-answer rate varies with density. Plain–thinking comparisons use the same graphs; we call the modes equivalent only when the 90% interval for their difference lies within ± 3 points. A non-significant effect remains unresolved at the study’s detectable effect sizes (Appendix A).

Evidence of graph use. Degree and neighbor-set questions target the same node. Their joint correctness tests whether a degree gain accompanies correct identification of that node’s neighbors; it does not establish the model’s internal procedure. Rule-based readers identify direct-answer and neighbor-list response patterns, primer conflicts, and graph claims. For correct, finished cycle-check answers, we distinguish a cited cycle whose steps are all edges, a claimed cycle containing a nonexistent edge, a walk that is not a cycle, and an answer without an explicit cycle. Within the last category, we also identify valid arguments from edge count: a

40-node graph with at least 40 edges must contain a cycle. Conflict detection was compared with blind hand labels; the other readers were spot-checked by an LLM under human supervision (Appendix A).

Follow-ups. We repeat Qwen3-1.7B node-degree tests on 400 graphs per density across all seven densities and run fresh-seed and 20–160-node checks at mean degrees 8 and 16; the size grid also includes Qwen3-8B.

4 Results

4.1 What the primers make available

The distinction between supplying an answer and supplying other structural information organizes the results (Appendix Figure 2). From stated degrees alone, the graph-blind solver recovers node degree directly and edge count by halving the degree sum, both with 100% accuracy. Without degrees, its accuracy is at most 21.0% on degree and 10.4% on edge count, compared with 14.4% and 3.0% without a primer (Appendix Table 2). This baseline does not rule out indirect clues: on sparse graphs, a lookup fitted on separate graphs recovers degree from rounded clustering or RWSE values substantially better than the majority answer (Appendix B). Across the main sweep, the largest gains occur when degrees supply the answer or its terms; primers without degrees and the controls produce effects of either sign (Table 1).

4.2 Stated degrees can help or disrupt counting

The degree primer raises node-degree accuracy by 7.5 points for plain 1.7B and 8.8 for thinking 1.7B, but lowers it by 6.5 for plain 4B, whose unaided accuracy is 99.25% (Table 1). Its effect depends on the cost of counting from the incident list. Within plain 4B, it harms queries about the lowest-degree nodes and helps those about the highest-degree nodes (-11.8 and $+10.6$ points); at $p = .85$, where unaided accuracy falls to 33.0%, it gains 27.0 points (Appendix B). Stated degrees also supply every term needed for edge count, raising plain 4B’s main-sweep accuracy by 27.2 points.

A stated value still has to be matched to the queried node. Across seven densities, plain 1.7B answers only 43% of degree prompts correctly despite having the value in the primer. At $p = .50$, 62% of plain 4B’s responses have the direct-answer pattern rather than an explicit neighbor list; about a third of its incorrect lookups use a degree printed one sentence away (Appendix B). Its losses are concentrated on prompts where the response pattern changes. These observations support answer lookup as an explanation for the primer’s effect, without establishing the model’s internal procedure.

4.3 Primers also change answers they do not supply

Filler alone changes up to one fifth of an arm’s outcomes, showing that added text can matter without adding a statistic (Appendix B). Effects of clustering and RWSE are generally smaller and vary by task and model. On plain 4B’s edge-existence queries, clustering raises balanced accuracy by 3.0 points while RWSE lowers it by 6.5 (Appendix Table 5). Combining all three statistics can reverse the effect of degree alone: on plain 4B’s dense degree queries, degree gains 9.3 points but the combined primer loses 21.3 on the same graphs (Appendix B).

The largest unflagged main-sweep gain without degrees is the component sentence’s 9.2-point increase in plain 4B’s *exact* neighbor-set accuracy. Its set-F1 rises more modestly, from 0.987 to 0.998 (Appendix B). The sentence is identical across all sampled graphs from $p = .20$, yet responses restate the queried node’s incident line much more often with it than with no primer or filler (98.5%, 63.0%, and 21.8%). Filler moves this response pattern in

the opposite direction and lowers exact-match accuracy by 12.5 points. Clustering’s smaller gain on plain 1.7B degree queries replicates on additional low-density graphs, but is specific to that setting (Appendix C).

4.4 Can a primer close the gap to thinking?

Where thinking outperforms plain mode, primers usually narrow rather than close the gap (Table 8). The component sentence brings plain 4B’s neighbor-set exact match within the prespecified ± 3 -point equivalence margin of thinking 4B without a primer; filler also affects this task. The combined primer reduces plain 1.7B’s node-degree gap from 15.8 to 5.5 points. Under degree, its neighbor-set gap is not significant, but the interval extends beyond the equivalence margin. Clustering and RWSE close less of the gaps.

The degree primer puts plain 4B ahead of thinking 4B on edge count. That comparison is strongly affected by completion: thinking 4B is correct on 97.6% of the edge-count responses it finishes, while many responses exhaust their token budget. Stated degrees also make edge count directly computable.

4.5 Primers in thinking mode

Thinking 4B is near ceiling on degree, neighbor sets, and edge existence; every primer changes these accuracies by at most 2.5 points. Thinking 1.7B gains from stated degrees but often lists neighbors and checks its count against the printed value. When the two conflict, it ends with the stated value in about two thirds of cases. This additional work raises its node-degree truncation rate from 1.1% to 7.6% across seven densities. Truncation also accounts for most changed answers in several thinking-4B comparisons (Table 7), so some apparent primer effects reflect whether a response finishes.

4.6 Do correct answers have graph support?

Clustering and RWSE do not significantly improve the joint correctness of degree and neighbor-set answers for the same node in any arm (Appendix B). The degree primer lowers this measure for plain 4B by 9.2 points, while the component sentence raises it by 9.2. These comparisons test whether a degree gain accompanies identification of the queried node’s neighbors; they cannot determine the model’s internal computation.

Every sampled graph contains a cycle, so cycle-check accuracy alone is uninformative about detecting cycles. We instead examine the support in correct, finished “yes” responses (Table 6). Without a primer, valid cited cycles or edge-count arguments support 34.2% and 58.5% of all prompts for plain 1.7B and plain 4B, respectively. Among plain 1.7B’s correct, finished responses, 43.5% cite an invented cycle containing a nonexistent edge. The degree primer raises its share of prompts with valid support to 50.2%, although cycle-check accuracy falls by 17.5 points. It also shifts plain-model responses toward valid arguments from edge count, which do not require naming a cycle. Thus, a correct “yes” can be accompanied by a valid proof, an invalid claimed cycle, or no explicit cycle; the label alone does not distinguish them.

5 Discussion

Structural statistics serve a different role here than in graph models. Random-walk encodings can distinguish structures that some message-passing networks cannot represent (Xu et al., 2019; Dwivedi et al., 2022). In our setting, the incident encoding already contains every edge, but the language model must locate and compute from them. Stated degree can bypass that computation for degree and edge-count queries. It also introduces a lookup that can fail when the model selects the wrong node’s sentence. Whether it helps therefore depends partly on the model’s unaided accuracy and its ability to use the stated value.

Clustering and RWSE do not directly answer the studied queries. Their mixed effects, together with changes caused by filler and the near-constant component sentence, are consistent with primers changing how models use the prompt. The independent-edge graphs provide limited opportunity for these statistics to convey task-relevant structure beyond degree and density; the results do not establish how they would behave on graph families where triangles or return patterns are more informative.

Finally, answer accuracy and the stated justification measure different things. Joint degree and neighbor-set correctness gives evidence about use of the incident line, while cycle checks reveal whether a correct “yes” is supported by a valid graph argument. Neither reveals the model’s internal computation. Interventions that alter a stated value while preserving the graph would test reliance on the primer more directly.

6 Future Work

Corrupting or omitting queried-node degrees while holding graphs fixed would test reliance on the primer versus counting. Shuffled labels and primer order, and a length- and position-matched neutral substitute for the component sentence, would test presentation effects. Acyclic graphs and graph families varying in triangle counts and return patterns at matched size and density would test cycle detection and structural primers. Other model families and token budgets would assess generality.

7 Conclusion

We asked whether explicit structural statistics improve a black-box language model’s execution of queries over a textual graph. In paired GraphQA tests, stated degrees produced the clearest gains, chiefly where they supplied an answer the model counted unreliably; they could also disrupt accurate counting. Clustering and random-walk returns produced smaller, task-dependent effects without a consistent gain in joint degree and neighbor-set correctness. Controls and graph-checked responses showed why accuracy alone is insufficient: added text can change answers, and even a correct cycle label can cite a nonexistent edge. The value of a textual statistic depends on the query, what the statistic gives away, and what the model can already compute.

Limitations

We test two Qwen3 models, one incident encoding, fixed prompts, and mostly 40-node independent-edge graphs; results may differ where triangles or return patterns are more informative. All sampled graphs contain cycles, so cycle analysis evaluates support for “yes” answers, not cycle discrimination. The filler only roughly matches clustering’s length, and component count supplies structural information. The graph-blind solver cannot exclude other clues. We did not test incorrect or shuffled statistics or an instruction-only control; node position and identifier length are confounded. Fixed budgets cause substantial truncation in some thinking-mode cells, and reruns can differ. Response categories describe behavior, not internal computation: only conflict readers were checked against blind hand labels; the others received supervised spot checks.

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A Method Details

Primer statistics. For the undirected adjacency matrix A , let $d_v = \sum_u A_{vu}$ and $P_{vu} = A_{vu}/d_v$ when $d_v > 0$ (a zero row otherwise). If t_v is the number of edges among v ’s neighbors, local clustering is $c_v = 2t_v/[d_v(d_v - 1)]$, set to zero for $d_v < 2$. The two return features are $r_v^{(k)} = (P^k)_{vv}$ for $k \in \{2, 3\}$. Floats are printed to two decimals, with one sentence per node in node order (Table 3).

Feature choice. Degree and random-walk encodings are used in graph Transformers (Rampásek et al., 2022); clustering records edges among a node’s neighbors. We use return steps 2 and 3: step 1 is zero in a simple graph, while step 3 reflects triangles. We omit Laplacian coordinates because their signs and degenerate eigenspace bases require a convention (Dwivedi et al., 2022). Component count is a global control, though it is informative in principle: a graph with n nodes, m edges and c components is acyclic exactly when $m = n - c$.

Models, budgets and inference. We run the Qwen3 checkpoints with `transformers`, greedy decoding and one response per prompt. The output limit is 8,192 tokens, except 2,048 for plain

arms at $p \geq .65$; none of these shorter-budget cells truncates above 2.5%. The main sweep and extension comprise 84,000 responses on 700 graphs. In one 100-graph cell the median smallest detectable paired effect is 7.4 points, and 13.7 where side information meets intermediate accuracy. Under one BH correction over all 264 tests, every bold cell of Table 1 but one and all but three ^f marks stay significant. Even greedy reruns change correctness on 5.3% of identical prompts. Paired effects therefore compare net correct shares; the number of responses that switch outcome is a separate descriptive measure.

Graph-blind solver. It receives the printed primer but neither the incident encoding nor the queried graph. It uses 16 exact rules, one heuristic and eight fitted rules; fitting and evaluation use disjoint graphs. Its score is a constructive shortcut, not an upper bound on information in the primer (Table 2).

Primer	Degree	Neighbors	Edge count	Edge exists
none	14.4	0.0	3.0	73.5
degree	100.0	0.1	100.0	74.6
clustering	14.4	0.0	10.4	72.7
RWSE	21.0	0.1	3.0	55.4
all	100.0	0.8	100.0	74.1
components	14.4	0.0	3.0	73.5
filler	14.4	0.0	3.0	73.5

Table 2: Graph-blind solver accuracy (%) on the four variable-answer tasks, averaged over seven densities ($p = .10-.85$, 40 nodes). It sees only the primer at evaluation; fitted rules use separate graphs. The RWSE rule’s density threshold fails at $p \geq .65$, where its edge-existence scores are 38.0, 48.0 and 12.0%.

Primer	Mean chars	Information
degree	891	Degree for each node
clustering	1,629	Local clustering for each node
RWSE	2,949	Two rounded return values per node
all	4,691	Degree, clustering and RWSE
components	37	Number of connected components
filler	1,829	Node names, no graph statistic

Table 3: Mean preamble length in characters over the main-sweep graphs.

Response readers. A *look-up* answer does not restate the queried node’s neighbor list; the list detector requires at least five neighbors on separate lines. Cycle and edge readers check stated graph

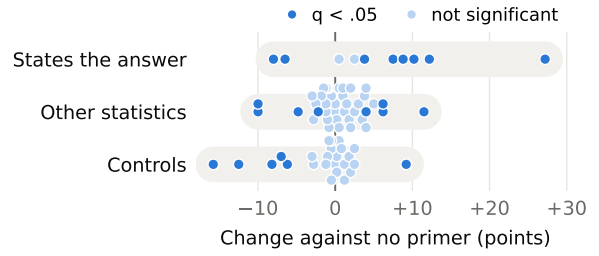


Figure 2: Unflagged main-sweep effects from Table 1, grouped by whether the primer supplies the answer, gives another statistic, or is a control. Each dot is one arm-task-primer cell; filled dots have BH $q < .05$ against no primer. The answer-stating group contains degree and all on node-degree and edge-count questions.

steps against the incident encoding. These behavioral labels do not establish the model’s internal computation. Only the conflict readers were compared with blind hand labels; the others received supervised spot checks.

Scoring. The answer extractor parses 99.84% of responses. Neighbor-set exact match is primary; set-F1 gives partial credit and moves in the same direction in three of five large contrasts, but by only 0.11–0.17 as much. Edge-existence balanced accuracy uses paired permutation tests with BH adjustment over the six conditions within each arm.

B Detailed Results

Tables 4 and 7 extend the main result with bootstrap intervals, truncated shares and the two constant-answer tasks. Figure 2 groups the main effects; Figures 3 and 4 show the density follow-up. The remaining tables separate edge-existence balance, cycle support and the plain-thinking gap.

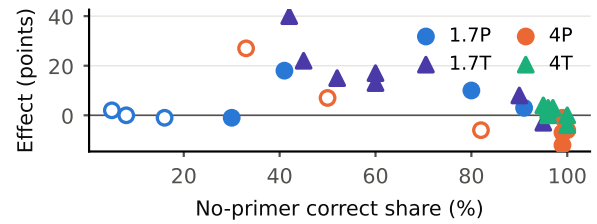


Figure 3: The degree primer’s effect on node degree against the no-primer correct share, per arm and density. Hollow: plain arm at $p \geq .65$ (smaller budget).

Stated answers (§4.2). The effect varies with unaided accuracy (Figure 3). In node-degree cells answered 25–50% and 50–75% correctly without them, answer-stating primers gain 16.1 and 12.9 points; at 75–90% they lose 5.8. These bands hold

Task	Arm	degree	clustering	RWSE	all	components	filler
Degree	1.7P	+7.5 [+3.0, +11.8]	+3.0 [-1.5, +7.5]	+4.0 [-0.5, +8.5]	+10.2 [+5.2, +15.2]	+1.8 [-2.2, +5.8]	-0.5 [-4.2, +3.5]
	1.7T	+8.8 [+4.2, +13.2]	+1.5 [-2.5, +5.5]	+2.5 [-2.0, +6.8]	+12.2 [+8.0, +16.5]	-1.0 [-4.2, +2.5]	-3.0 [-7.0, +0.8]
	4P	-6.5 [-9.2, -4.0]	-0.5 [-2.0, +0.8]	-2.2 [-4.2, -0.5]	-8.0 [-11.0, -5.2]	+0.2 [+0.0, +0.8]	-0.5 [-2.0, +1.0]
	4T	+2.5 [+0.8, +4.2]	+1.8 [-0.2, +3.8]	-1.2 [-3.8, +1.0]	+0.5 [-1.5, +2.8]	+0.2 [-1.8, +2.5]	+2.0 [+0.5, +3.8]
Neighbors	1.7P	+3.5 [-0.5, +7.2]	+1.2 [-3.0, +5.5]	+1.0 [-3.2, +5.2]	+2.0 [-2.5, +6.5]	+0.8 [-2.5, +4.5]	-6.2 [-10.5, -2.2]
	1.7T	+6.2 [+2.8, +10.0]	+2.0 [-2.0, +6.0]	+0.2 [-3.5, +4.0]	+6.2 [+2.5, +10.0]	+2.5 [-0.8, +6.0]	+2.5 [-1.0, +6.0]
	4P	-3.0 [-7.0, +1.0]	+1.0 [-2.8, +4.8]	+0.0 [-3.8, +4.0]	-2.8 [-7.0, +1.8]	+9.2 [+6.0, +12.8]	-12.5 [-17.2, -7.8]
	4T	+0.5 [-2.2, +3.2]	-0.8 [-3.2, +1.8]	+0.5 [-1.8, +3.0]	-1.2 [-3.8, +1.2]	+0.5 [-1.8, +2.8]	+1.2 [-1.0, +3.5]
Edge count	1.7P	+3.0 [†] [+0.8, +5.2]	-0.5 [†] [-2.0, +1.0]	-0.8 [†] [-2.0, +0.5]	+0.8 [†] [-1.0, +2.5]	-1.0 [†] [-2.2, +0.0]	-0.5 [†] [-1.8, +0.8]
	1.7T	-3.2 [†] [-7.8, +1.2]	-1.2 [†] [-5.5, +3.0]	-3.2 [†] [-7.2, +0.5]	+5.0 [†] [+0.2, +10.0]	-0.8 [†] [-5.0, +3.8]	-1.5 [†] [-5.5, +2.5]
	4P	+27.2 [+22.8, +32.0]	-1.2 [-3.0, +0.2]	-1.5 [-3.0, -0.2]	+3.8 [+1.2, +6.2]	-1.2 [-2.8, +0.0]	+1.2 [-0.8, +3.0]
	4T	+8.5 [†] [+3.2, +14.2]	+24.8 [†] [+19.5, +30.2]	+10.2 [†] [+5.5, +15.0]	+1.8 [†] [-3.5, +7.2]	+24.8 [†] [+19.8, +30.0]	+12.2 [†] [+7.2, +17.0]
Edge exists	1.7P	+4.0 [-0.2, +8.2]	+3.8 [-0.5, +8.0]	+5.0 [+0.2, +9.5]	+11.5 [+6.2, +16.5]	-8.2 [-11.8, -4.8]	-15.8 [-20.0, -11.8]
	1.7T	-2.5 [-4.5, -0.8]	-1.0 [-2.8, +0.5]	-1.2 [-3.0, +0.5]	-1.8 [-3.8, +0.0]	-0.8 [-2.5, +0.8]	-0.5 [-2.0, +1.0]
	4P	-4.8 [-7.8, -1.5]	+4.0 [+1.0, +7.0]	-10.0 [-13.8, -6.2]	-10.0 [-14.0, -6.0]	-2.8 [-5.8, +0.2]	-7.0 [-10.5, -3.5]
	4T	+0.0 [-0.8, +0.8]	-0.8 [-1.8, +0.0]	-1.2 [-2.5, +0.0]	-1.0 [-2.2, +0.0]	-0.5 [-1.5, +0.5]	+0.0 [+0.0, +0.0]
Node count	1.7P	+6.8 [+3.2, +10.5]	+67.2 [+64.5, +70.0]	+2.0 [-3.2, +6.8]	+47.0 [+42.5, +51.2]	+21.2 [+16.2, +26.2]	+39.8 [+36.0, +43.5]
	1.7T	+5.5 [+2.5, +8.5]	+9.0 [+6.5, +11.8]	+9.0 [+6.5, +11.8]	+9.0 [+6.5, +11.8]	+1.5 [-2.2, +5.0]	+9.0 [+6.5, +11.8]
	4P	+8.0 [+5.2, +10.8]	+8.8 [+6.0, +11.5]	+9.2 [+6.5, +12.0]	+9.8 [+7.2, +12.5]	+9.0 [+6.5, +11.8]	+9.2 [+6.8, +11.8]
	4T	+0.2 [-0.5, +1.2]	+0.2 [-0.5, +1.2]	+0.2 [-0.5, +1.2]	+0.5 [+0.0, +1.2]	+0.0 [-1.0, +1.0]	+0.5 [+0.0, +1.2]
Cycle	1.7P	-17.5 [†] [-23.0, -12.2]	+8.5 [+5.8, +11.2]	-0.2 [-4.2, +3.5]	+7.0 [+4.0, +10.2]	+2.0 [-1.5, +5.3]	+3.0 [-0.8, +6.8]
	1.7T	-21.8 [†] [-26.5, -17.5]	-0.5 [-2.8, +1.8]	+2.2 [+0.5, +4.0]	-0.8 [-3.2, +1.8]	-7.5 [-11.0, -4.2]	+3.0 [+1.5, +4.8]
	4P	-2.2 [-4.2, -0.5]	+0.5 [-0.5, +1.8]	-0.2 [-1.8, +1.2]	-0.2 [-1.8, +1.0]	-2.0 [-4.0, -0.2]	-0.5 [-2.0, +0.8]
	4T	-5.5 [-8.0, -3.0]	+0.0 [-1.2, +1.2]	+0.2 [-0.8, +1.5]	-0.5 [-2.0, +0.8]	-3.2 [-5.2, -1.2]	+0.5 [-0.5, +1.5]

Table 4: Every main-sweep effect against no primer (points) with its 95% bootstrap interval over graphs, 400 paired graphs per cell ($p \leq .50$). [†]: at least 15% truncation on either side. Node count (always 40) and cycle check (always yes) cannot measure reading the graph at 40 nodes. P/T: plain/thinking.

on held-out halves of the graphs and with the truncation flag at 10% or 50% (at least +12.5 in every case). An effect mostly follows how well the model does unaided. At a fixed accuracy without a primer, density adds little ($p = .28$ for answer-stating primers, $p = .85$ for side statistics), except that side statistics' effect on node degree falls with density ($p = .006$). One model falls on both sides. The degree primer costs plain 4B 12.0 points at $p = .50$, where it answers 99.0% correctly unaided, and gains 27.0 at $p = .85$, where it answers 33.0% (both $q < .01$). Stated answers are not always taken: across all seven densities, where the solver scores 100%, node-degree accuracy under the degree primer is 43% and 81% for plain 1.7B and plain 4B, and 79% and 98% with thinking. At high density, where plain 1.7B is weakest, the stated degree barely helps it (at most 2 points). At $p = .85$ it still answers 39, the degree of a node joined to all others, on 70.8% of finished responses. Plain 4B's wrong look-ups copy a nearby line: 17 of 57 equal the degree printed one sentence away, against 10.4 expected by chance ($p = .017$). Adding clustering and RWSE to the same degrees costs dense plain 4B the most (Figure 4): pooled over $p \geq .65$, degree alone gains 9.3 points ($q = .03$) and all three statistics lose 21.3 ($q < .001$).

Filler and fragile questions (§4.3). Filler changes 6.2–18.6% of outcomes per arm. A primer changes 38.5–69.1% of the questions filler changes and 3.1–15.2% of the rest. Some questions are fragile: on plain 1.7B's node degree, 24 of 400

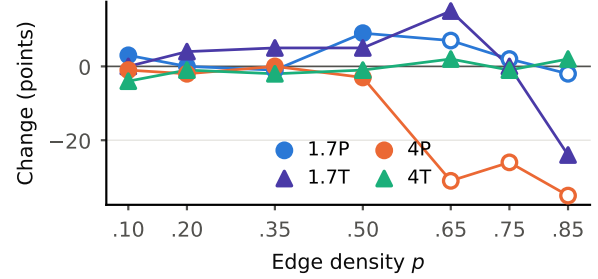


Figure 4: What adding clustering and RWSE to the same degrees changes on node degree (all three statistics minus the degree primer, points), per arm and density. Hollow as in Figure 3.

change outcome when the same prompt is generated again. Every condition changes them far more often than the rest (45.8–66.7% against 14.4–26.1%, $q \leq .018$). Yet they carry only 11 to 16 of each condition's 65 to 113 changes, a lower bound on chance with only two generations per prompt. Where accuracy without a primer is 25–75%, components, clustering and RWSE average +1.6 points while filler averages -6.9 on the same cells.

Side statistics (§4.3). On edge existence, every primer that moves balanced accuracy does so in a plain arm (bold in Table 5); clustering and RWSE on plain 4B have $q \leq .018$, and all raises plain 1.7B's by 7.8 points and lowers plain 4B's by 6.5. Plain 1.7B's false alarms rise steeply with the number of neighbors a non-edge pair shares, from 1.7% with none to 65.9% with four or more, and plain 4B's from 0.2% to 38.8%. In $G(n, p)$ shared neighbors say nothing about whether a pair is joined, so

Arm	Primer	Raw	Balanced	Yes
1.7P	none	70.50	79.86	56.2
	degree	74.50	82.30	51.7
	clustering	74.25	82.42	52.5
	RWSE	75.50	82.68	50.1
	all	82.00	87.71	44.8
	components	62.25	74.23	64.5
1.7T	filler	54.75	69.11	72.0
	none	99.00	99.02	27.1
	degree	96.50	97.61	29.9
	clustering	98.00	98.63	28.6
	RWSE	97.75	98.17	28.3
	all	97.25	98.12	29.5
4P	components	98.25	98.81	28.3
	filler	98.50	98.68	27.8
	none	90.50	93.22	35.7
	degree	85.75	90.27	40.7
	clustering	94.50	96.25	32.3
	RWSE	80.50	86.69	46.3
4T	all	80.50	86.69	46.1
	components	87.75	91.64	39.0
	filler	83.50	88.74	43.2
	none	99.75	99.83	26.8
	degree	99.75	99.83	26.8
	clustering	99.00	99.32	27.0
	RWSE	98.50	98.68	26.9
	all	98.75	99.15	27.1
	components	99.25	99.49	27.0
	filler	99.75	99.83	26.8

Table 5: Edge existence, main sweep (400 prompts per row): raw and balanced accuracy over all prompts and the share of finished answers that say yes, all in %. Truncation is in Table 7. Bold: balanced accuracy differs from the same arm without a primer (paired permutation, BH $q < .05$ within arm).

this is consistent with mistaking nearby ids for adjacency. Under clustering, plain 1.7B’s responses explicitly claiming a non-edge is an edge fall from 13.7% to 4.4%, while its false alarms fall only from 40.3% to 35.2%: stated reasons move more than decisions. The component sentence raises plain 4B’s neighbor-set Set-F1 from 0.987 to 0.998, and responses that restate the queried node’s line are right 95%, 92% and 84% of the time with it, with no primer and with filler. At high density the printed statistics vary less within a graph: clustering values have SD 0.021 at $p = .65$, and rounded RWSE takes 3.30 distinct value pairs per graph at $p = .50$, against 12.50 distinct degrees. Clustering still gains 11.3 on dense plain 4B node degree with similar rates of neighbor enumeration, making presentation a plausible contributor.

Against the thinking mode (§4.4). Equivalence in Table 8 rests on 90% intervals. The component sentence’s gap on plain 4B’s neighbor sets lies in $[-2.0, +2.5]$, inside the ± 3 -point margin. The de-

Arm	Primer	Correct “yes” (n)	Cycle named			No cycle	
			Real	Invented	Not a cycle	All	Edge count
1.7P	none	363	36.9	43.5	16.0	3.6	0.8
	degree	293	13.0	20.5	1.4	65.2	55.6
	clustering	397	39.5	48.1	7.8	4.5	1.0
	RWSE	362	33.7	40.3	19.9	6.1	1.7
	all	391	38.4	50.4	7.9	3.3	1.3
	components	371	27.8	40.4	25.3	6.5	4.3
1.7T	filler	375	33.3	55.5	9.6	1.6	0.0
	none	388	79.9	17.8	1.0	1.3	0.5
	degree	301	61.5	29.6	0.0	9.0	9.0
	clustering	386	80.6	19.2	0.0	0.3	0.3
	RWSE	397	68.0	20.7	10.6	0.8	0.5
	all	385	73.0	21.8	3.1	2.1	1.8
4P	components	358	74.9	22.6	0.0	2.5	2.0
	filler	400	77.2	19.8	2.5	0.5	0.0
	none	396	34.1	23.0	6.8	36.1	25.0
	degree	387	13.4	3.9	2.1	80.6	65.6
	clustering	398	48.7	31.2	1.0	19.1	5.8
	RWSE	395	33.2	20.8	38.7	7.3	0.5
4T	all	395	45.1	33.4	6.6	14.9	0.8
	components	388	39.2	18.6	3.4	38.9	26.5
	filler	394	44.2	28.2	5.3	22.3	2.3
	none	397	96.5	3.3	0.0	0.3	0.0
	degree	375	90.9	5.1	0.0	4.0	2.7
	clustering	397	95.5	4.5	0.0	0.0	0.0
	RWSE	398	55.0	2.5	42.5	0.0	0.0
	all	395	92.7	6.3	0.0	1.0	1.0
	components	384	92.2	4.4	0.0	3.4	2.9
	filler	399	94.2	5.0	0.5	0.3	0.0

Table 6: What the correct cycle-check answers rest on (main sweep; every correct answer is “yes”). *Correct (n)*: count of correct, finished answers; the other columns are % of those n , and Real, Invented, Not a cycle and All sum to 100. *Real*: every step of the cycle is an edge; *Invented*: a step is not an edge; *Not a cycle*: a length-2 or retraced walk. *Edge count*: the answers under All that argue from the number of edges, identified by their wording. The valid arguments are Real and Edge count, which is valid because 40 nodes with at least 40 edges must contain a cycle. Bold (Real, Invented only): differs from the same model without a primer (paired, BH $q < .05$).

gree primer’s gap on plain 1.7B’s neighbor sets is not significant ($q = .24$), but its interval reaches $+6.2$, so the two are not shown to be level.

Thinking mode (§4.5). Thinking-mode responses often check stated values against their own count. Under the degree primer, thinking 1.7B still lists the neighbors in 90% of responses at $p \geq .35$; when their count conflicts with the stated value, thinking 1.7B and thinking 4B end on the stated value in 66% and 93% of cases. In the follow-ups, clustering gains thinking 1.7B 2.5 points at $p \leq .50$ ($q = .043$) and loses 5.4 at $p \geq .65$ ($q = .0015$), where filler loses 11.4. Of the answers a condition

Task	Arm	none	degree	clustering	RWSE	all	components	filler
Degree	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.75	+4.5	-0.5	+0.0	-0.5	-0.2	-0.5
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	2.75	-2.5	-2.0	-0.5	-1.0	-0.5	-1.8
Neighbors	1.7P	0.00	+0.0	+0.0	+0.0	+0.2	+0.0	+0.0
	1.7T	1.25	+0.2	-1.0	-0.8	-0.5	-0.2	-0.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	3.25	-0.2	+1.2	-0.2	+1.0	+0.2	-0.5
Edge count	1.7P	29.75	-20.2	-11.8	-11.8	-1.8	-2.5	-3.5
	1.7T	81.75	-0.8	+0.8	+5.0	-12.2	+0.5	+1.2
	4P	0.00	+1.8	+0.2	+0.5	+0.8	+0.8	+0.0
	4T	79.50	-20.8	-26.0	-12.2	-23.8	-25.0	-13.0
Edge exists	1.7P	0.00	+0.0	+0.0	+0.2	+0.0	+0.0	+0.0
	1.7T	0.25	+0.2	+0.0	+0.0	-0.2	+0.0	-0.2
	4P	0.00	+0.5	+0.0	+0.0	+0.2	+0.0	+0.0
	4T	0.25	+0.0	+0.8	+1.2	+1.0	+0.5	+0.0
Node count	1.7P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	1.7T	7.75	-5.0	-7.8	-7.8	-7.8	-0.5	-7.8
	4P	0.00	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	4T	0.00	+0.2	+0.2	+0.2	+0.0	+0.0	+0.0
Cycle	1.7P	9.00	+8.2	-8.2	-7.2	-8.0	-2.5	-2.8
	1.7T	3.00	+21.2	+0.5	-2.2	+0.8	+7.5	-3.0
	4P	0.75	+0.8	-0.2	+0.5	-0.2	+0.8	+0.5
	4T	0.75	+4.8	+0.0	-0.2	+0.5	+3.2	-0.5

Table 7: Complete main-sweep truncated shares: no-primer baseline (percent of all prompts) and paired changes (percentage points), for the same queries as Table 4. Each of its cells’ *wrong* share equals 100% minus its correct and truncated shares. P/T: plain/thinking; components and filler are diagnostics. No significance marks are assigned to truncation contrasts.

Task	Model	Gap without primer	Best primer	Gap after primer	Gap with CI/RW.
Degree	1.7B	+15.8	all	+5.5	+11.8
	4B	-2.2	components	-2.5	-1.8
Neighbors	1.7B	+6.2	degree	+2.8	+5.0
	4B	+9.5	components	+0.2	+8.5
Edge count [†]	1.7B	+14.5	degree	+11.5	+15.0
	4B	+18.0	degree	-9.2	+19.2
Edge exists	1.7B	+28.5	all	+17.0	+23.5
	4B	+9.2	clustering	+5.2	+5.2

Table 8: Gap in correct share (points): the same model thinking without a primer minus plain under no primer, its held-out-selected best primer, or the better of clustering and RWSE. Negative values put plain ahead. Bold: paired BH $q < .05$; [†]: thinking edge count truncates frequently (Table 7). Selection and equivalence details are in the text.

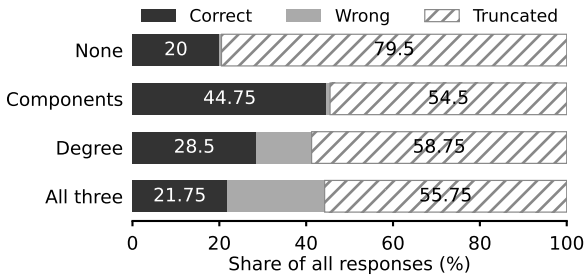


Figure 5: Thinking 4B on edge count: correct, wrong and truncated shares of *all* responses. Conditions from top: none, components, degree, all three.

changes, 44.4–64.5% for thinking 1.7B and 83.3–95.2% for thinking 4B involve a response that ran out of tokens; Figure 5 shows this for thinking 4B’s

edge count.

Graph support (§4.6). Neither clustering nor RWSE significantly changes joint degree–neighbor correctness in any arm ($q \geq .16$). Plain 4B’s is 85.75% with no primer, 76.50% with degree and 95.00% with the component sentence; the degree primer raises plain 1.7B’s from 54.00% to 62.50%. Table 6 classifies every correct, finished cycle-check answer by the argument it cites. For example, on one $p = .20$ graph plain 1.7B without a primer answers with $0 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 14 \rightarrow 13 \rightarrow 15 \rightarrow 16 \rightarrow 18 \rightarrow \dots \rightarrow 38 \rightarrow 39 \rightarrow 0$, which climbs the ids almost in order; steps such as $15 \rightarrow 16$, $21 \rightarrow 25$ and $38 \rightarrow 39$ are not edges. With thinking and no primer, valid arguments back 78.0% and 95.8% of all prompts. The degree primer raises plain 4B’s valid share from 58.5% to 76.5%. Truncation accounts for 21.2 of the 21.8 points it costs thinking 1.7B but only 8.2 of the 17.5 it costs plain 1.7B (Tables 4 and 7). RWSE, all three statistics and filler lower plain 4B’s valid share to 33.2%, 45.2% and 45.8% ($q < .05$).

Node count. Because the encoding lists all 40 ids, this constant-answer task cannot establish graph reading. Plain 1.7B answers 39, the last id, on 68.5% of finished responses without a primer and on 1.2% under clustering, hence Table 4’s large node-count effects.

Test	Gain	p or q
Screened graphs (700)	+1.29	$p = .47$
Unscreened graphs (2,100)	+2.1	$p = .031$
$p \leq .50$ (1,600 pairs)	+3.8	$q = .0035$
Fresh seed	+4.2	$p = .00055$ (.022 corr.)
Against filler	+3.2	$q = .014$
Fixed mean degree (3,200)	+6.2	$q < .001$
$p \geq .65$ (1,200 pairs)	−0.7	$q = .57$
Qwen3-8B (3,200)	−1.0	$q = .05$

Table 9: Change in node-degree accuracy (points) for plain Qwen3-1.7B, except the Qwen3-8B row. Comparisons use no primer unless labeled against filler. *Screened*: graphs reused from the discovery screen; *fixed mean degree*: the 20–160-node grid; *corr.*: Bonferroni correction over screened cells. The p and q columns retain their distinct adjustments.

C Pilot and the Clustering Screen

Pilot. On the published GraphQA graphs (5–19 nodes), Gemma 4 E4B/12B (Gemma Team, 2026) and Qwen3-8B/14B, in both modes, were near ceiling without a primer on most tasks. Edge count remained difficult. These pilot results motivated the 40-node graphs and smaller models; they are not pooled with the main sweep.

The clustering effect. Clustering on plain 1.7B’s node degree emerged from a 40-cell screen, so Table 9 tests it on graphs the screen did not see, on a fresh seed, against filler, on 20–160-node graphs at fixed mean degree, at high density and on Qwen3-8B. The low-density gain appears on unscreened graphs, a fresh seed and the fixed-mean-degree grid, but is not detected at high density or in Qwen3-8B (Table 9). The screened-graph row is descriptive, because it reuses graphs from the initial search. No sampled response explicitly invokes clustering, and the gain does not vary detectably with the queried node’s coefficient ($p = .50$). These observations motivate a position-based explanation but do not prove one. The clustering contrast is +5.01, +0.75 and −0.11 points for queried nodes in the first, middle and last third of the list, pooled across seven densities. A regression adjusting for degree, clustering and density estimates 1.74 points less gain per ten later positions ($p = .018$); filler and component controls show no corresponding gradient. The clustering format resolves 224 prior neighbor-list copying errors and introduces 176 new ones. These associations do not isolate the cause of the gain; shuffling node order would test it directly. Components and filler also cut the neighbors the model misses (0.94 and 1.05 per response at $p = .35$ –.50, against 0.86 with clustering and 1.49 with none).